

Aman Gupta

Mobile: +91-9540010070 | Email: aman.g191193@gmail.com | GitHub: github.com/kaashgarg19
LinkedIn: linkedin.com/in/amangupta1911 | Portfolio: kaashgarg19.github.io/kaashgarg19/ | ORCID: orcid.org/0009-0007-4112-9993

Research Profile

Computer Science graduate with an MSc in Advanced Computer Science (Distinction) and a research background in environmental IoT sensor analytics and machine learning. My MSc dissertation investigated the prediction of human movement near IoT devices. I am now extending that work through reproducible experiments focused on rare-event prediction, temporal evaluation and generalisation across sensing devices.

Research Experience

IoT Human Movement Prediction — MSc Dissertation & Research Extension | 2021 – Present

- Investigated whether environmental IoT sensor measurements could be used to predict human movement near IoT devices as part of my MSc dissertation.
- Reconstructed and documented the original MSc methodology, dataset, modelling approach and results as a reproducible research record.
- Extended the analysis beyond the original work to examine random versus chronological evaluation, device-level generalisation, feature ablation and decision-threshold sensitivity.
- Evaluated Logistic Regression, Random Forest and Histogram Gradient Boosting using metrics including average precision, precision, recall and F1-score because movement events are extremely rare.
- Documented the dataset, codebook, methodology, experimental protocol, reproducibility requirements and research results in a public GitHub research repository.

Research Interests

Machine learning for IoT sensing; human movement prediction; sensor-data analytics; rare-event prediction; temporal generalisation; distribution shift; model evaluation; reproducible machine-learning workflows.

Education

Birmingham City University, UK

Master of Science in Advanced Computer Science (Distinction)

January 2020 – June 2021

- Dissertation: IoT Analytics for Predicting Human Movement near IoT Devices

Maharshi Dayanand University, India

Bachelor of Technology in Computer Science Engineering (First Division)

August 2011 – May 2015

Technical Skills

- **Programming & Data:** Python, SQL, MySQL, Pandas, NumPy, JavaScript, C#
- **Machine Learning:** Scikit-learn, Classification, Regression, Predictive Modelling, Model Evaluation, Imbalanced Data
- **Analytics & Visualisation:** Power BI, Microsoft Excel, Jupyter, Matplotlib, Exploratory Data Analysis
- **Tools & Platforms:** Git, GitHub, SQL Server, Macromatix, Microsoft Office Suite

Selected Projects

- **Bike Sales SQL Analysis** — SQL joins, aggregation, revenue calculations, customer order analysis and relational sales data.
- **Amazon Prime Data Analysis** — Python, pandas and exploratory data analysis to examine dataset structure, patterns and trends.
- **Stock Market Portfolio Analysis** — Historical market data, returns, risk measures and portfolio calculations.
- **Data Science Coursework** — Python-based data analysis and machine-learning techniques applied to practical coursework datasets.
- **Opportunity Dataset Team Project** — Team-based data preparation, analysis, visualisation and reporting.

Professional Experience

Concentrix — Advisor I, Customer Service | November 2025 – Present

- Review and resolve customer cases while following documented processes and service policies.
- Identify cases requiring escalation and maintain clear and accurate case records.
- Use structured problem-solving and communication skills to resolve customer issues efficiently.

Team Member / Trainee Team Leader — KFC, Yum Brands | September 2022 – November 2024

- Supported restaurant operations, customer service, stock management and team coordination in a high-volume environment.
- Progressed to Trainee Team Leader and supported day-to-day team supervision and operational activities.
- Used operational systems and procedures to support scheduling, inventory management and service delivery.
- Trained and supported new team members on operational procedures, software tools and safety requirements.

Device Testing Volunteer — Birmingham City University | March 2021 – May 2021

- Conducted benchmarking and usability tests on academic devices using PassMark.
- Provided feedback on hardware performance, compatibility and usability for academic environments.

Research Portfolio

- GitHub: github.com/kaashgarg19
- Research repository: github.com/kaashgarg19/kaashgarg19/tree/main/research/iot-human-movement-prediction
- Portfolio: kaashgarg19.github.io/kaashgarg19/
- ORCID: orcid.org/0009-0007-4112-9993